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Summary

Software Engineering Education researcher with experience designing adaptive learning systems, teaching large cohorts, and leading statewide educational data operations. Strong technical and pedagogical background with 12+ years of experience in education, analytics, and software development.

Education

- 2023 – present **Ph.D. Candidate in Computer Science, University of Alabama**
Advisor: Prof. Jeffrey Carver
Thesis: Learning Beyond the Lecture: Rethinking Software Engineering Education with Supplementary Materials in the Generative AI Era
Expected Completion: December 2026
- 2025 **M.S. in Computer Science, University of Alabama**
Comprehensive Exam: Effective Methods for Teaching Software Engineering
- 2023 **M.Sc. by Research in Computing Science, University of the South Pacific**
Supervisor: Dr. Vani Vasudevan
Thesis: Intelligent Education Management Information System (IEMIS) for Fiji
- 2008 **B.Ed. in Computer Science, University of Ibadan**
Minor: Mathematics
First Class (Honours); Best Graduating Student, Computer Science Department

Research Publications

Journal Articles

1. **L. Badru** and J. Carver, "Personalizing supplementary materials in software engineering: Impact of preferences, learning styles, and gender," *ACM Transactions on Computing Education*, vol. 26, no. 4, 2026. [DOI: 10.1145/3806053](#)
2. **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, "Mern stack web-based education management information systems for pacific island countries," *SN Computer Science*, vol. 4, no. 1, p. 70, 2022. [DOI: 10.1007/s42979-022-01437-0](#)
3. **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, "State of education management information system (emis) in pacific melanesia small island developing states (sids)," *Directions: Journal of Educational Studies*, vol. 35, no. 2, pp. 107–118, 2021.

Conference Proceedings

1. **L. Badru**, M. An, J. Stamper, and J. Carver, "Optimizing learning: A comparative study of adaptive experiments and randomization in a software engineering course," in *Proceedings of the Sixth Annual Workshop on A/B Testing and Platform-Enabled Learning Engineering (PELE), Learning@Scale 2025*, Palermo, Italy, 2025.

Preprints

1. **L. Badru** and J. Carver, *How undergraduate software engineering students navigate learning materials across contexts*, Under Review *ACM Transactions on Computing Education (TOCE)*, Jun. 2026.

- 2 L. Badru and J. Carver, *What students use and what works: A systematic review in software engineering education*, Submitted to SIGCSE, Jul. 2026.

Work-in-progress

- 1 L. Badru and J. Carver, *Personalized support: Evaluating a digital platform for supplementary materials in software engineering education*, Work in progress, 2026.

Employment History

- 2023 – present  **Research Assistant**, Department of Computer Science, University of Alabama
- Conduct research in software engineering education and learning analytics.
 - Designed adaptive learning experiments used across multiple universities
 - Contributed to ACM publications and data analysis workflows
- 2016 – 2022  **Senior Education Officer / Acting Head, State EMIS**, Ministry of Education, Science and Technology, Abeokuta, Nigeria
- Directed statewide EMIS operations supporting 1,500+ schools and 500,000+ students
 - Developed analytics tools that improved reporting accuracy and reduced processing time
 - Trained 1,000+ personnel as UNICEF Master Trainer
 - Contributed to Education Sector Plan (2017–2030)
 - Served as a technical expert for the Education Policy & Strategy Simulation (EPSSim) initiative.
- 2016 – 2019  **Visiting Lecturer (Secondment)**, D.S. Adegbenro ICT Polytechnic, Nigeria
- Taught computing, mathematics, and information systems to cohorts of up to 120 students
 - Designed assessments, tutorials, and practical exercises
 - Strengthened student understanding using structured progression and worked examples
- 2014 – 2016  **Education Officer (Secondment)**, Staff Development Center, Bureau of Establishment & Training, Ogun State, Nigeria
- Delivered training in computer applications and digital literacy for confidential secretaries and administrative staff.
- 2010 – 2014  **Programmer Analyst**, Teaching Service Commission, Abeokuta, Nigeria
- Provided technical support for educational data systems and contributed to ICT capacity-building initiatives.

Teaching Experience

- May, 2026

Workshop Facilitator, Tai Solarin Federal University of Education, Ijagun, Ogun State, Nigeria
- Designed and led a two-day virtual workshop on *Agentic Coding* for advanced Computer Science and Software Engineering students.
 - Created interactive activities and guided practice sessions to reinforce key concepts.
- 2016 – 2019

Visiting Lecturer, D.S. Adegbenro ICT Polytechnic, Ogun State, Nigeria
- Large-class teaching (up to 120 students)
 - Structured progression, worked examples, formative assessment
- 2014 – 2016

Education Officer, Staff Development Center, Bureau of Establishment & Training, Nigeria
- Delivered training in computer applications and digital literacy for adult learners and administrative staff.
 - Developed instructional materials and hands-on exercises tailored to diverse learner backgrounds.
- 2010 – 2026

Mentorship and Student Support
- Mentored over 100 high school and college students, supporting academic growth and career development.
 - Provided guidance on computing fundamentals, project work, and professional pathways.

Skills

Programming Languages	Python, JavaScript, PHP, React, Node.js
Artificial Intelligence	OpenAI API, Prompt Engineering, AI-driven Adaptive Quizzes
Learning Analytics	Personalization Models, Predictive Analytics, Behaviour Modelling, A/B Testing
Educational Technology	Adaptive Experimentation Platforms, Learning Tool Integration, Technology-enhanced Teaching
Research Methods	Experimental Design, Mixed Methods, Survey Design, Statistical Modelling
Software Engineering Education	Curriculum Design, Instructional Materials, Pedagogical Strategies, Learning Resource Evaluation
Data & Research Tools	Alteryx, Tableau, Pandas, NumPy, SciPy
Databases	MySQL, PostgreSQL, MongoDB
Teaching & Facilitation	Lecture Design, Assessment Development, Worked Examples, Problem-based Learning, Large-class Teaching

Miscellaneous Experience

Scholarly Contribution

- 2026  **Reviewer**, *Special Interest Group on Computer Science Education (SIGCSE)*. Review manuscripts in Computing Education Research (CER).
- 2022  **Reviewer**, *Education and Information Technologies* (Springer Nature). Reviewed manuscripts in educational technology and learning analytics.

Selected Industry & Development Projects

- 2026–present  **Security Geolocation Application**. Developing a security-focused geolocation application to support operational decision-making in Nigeria.
 - Implementing real-time location tracking, mapping, and secure data management.
 - Applying software engineering principles across requirements, architecture, and iterative development.
 - Building practical, location-aware tools for high-stakes public safety contexts.
- 2026  **Financial Management Information System (FMIS)**. Co-developed a web-based FMIS for thrift and cooperative societies in Nigeria.
 - Contributed to requirements analysis, system design, database modelling, and iterative development.
 - Strengthened practical experience in secure data handling and operational software systems.

Awards and Scholarships

- 2021  **Research Grant**, Queen Elizabeth Commonwealth Scholarship, United Kingdom (3,000 USD)
- 2020 – 2021  **Commonwealth Scholarship**, United Kingdom (value over 50,000 USD), University of the South Pacific, Fiji
- 2019  **Dean's Award for Academic Excellence & Global Leader Scholarship**, University of Bath, United Kingdom (value £10,000)
- 2009  **Best Graduating Student**, Department of Computer Science, Tai Solarin University of Education, Nigeria (value 500 USD)

Certifications

- 2021  **Predictive Analytics for Business**, Udacity (Alteryx & Tableau)
-  **Microsoft Azure Fundamentals**, Microsoft
-  **Ideas for a Better World: Leading Change Through Policymaking**, British Council

Mentorship

- 2010 – 2026  Advised and mentored over 100 senior secondary and university students, supporting academic and professional development.
- 2026  **Judge**, Alabama Robotics Contest, University of Alabama. Evaluated robot navigation accuracy, task completion, and engineering design.

Professional Memberships

-  **Association for Computing Machinery (ACM)**
-  **Institute of Electrical and Electronic Engineers (IEEE)**
-  **Special Interest Group on Computer Science Education (SIGCSE)**

References

Available on Request